

Lab 2: Performing a Vulnerability Assessment

In Lab 2, I used Nmap to scan my network, applying -O, -sS, and -sV for OS detection, stealth scans, and service identification. I conducted Nessus and Greenbone Security Manager scans to detect vulnerabilities, researched mitigation strategies using [ndv.nist.gov](https://nvd.nist.gov), and performed a penetration test on drsst.com to analyze security risks. Using Linux commands like ping and traceroute, I mapped network connections and identified open ports. I also scanned the Domain Controller 01 IP, ran a Nessus vulnerability scan, and compiled a security report. Challenges included interpreting Nmap and Nessus scan results, troubleshooting inconsistent port scan outputs, and ensuring proper penetration testing techniques without misconfiguring security settings. This lab helped me refine my skills in network scanning, vulnerability assessment, and penetration testing, all critical for cybersecurity.

Lab 3: Performing Packet Capture and Traffic Analysis

In Lab 3, I used Wireshark to capture and analyze network traffic. One of the biggest challenges was sorting through all the data to find specific details like FTP transfers, ICMP payloads, and SSH encryption. It was also tricky to tell the difference between encrypted and unencrypted traffic and understand how passive and active FTP connections worked. To start, I set up Wireshark and generated network activity by transferring files using FTP and SFTP. I then analyzed the captured packets, looking at encryption methods, destination ports, and traffic patterns. I used filters in Wireshark to focus on different protocols, like ICMP, SSH, and FTP, to pull out the most relevant information. I also looked at Wi-Fi traffic by connecting to SecureLabs-WiFi and analyzing details like the SSID and security settings. This lab helped me get better at finding and analyzing network traffic efficiently. I also learned more about how encrypted protocols, like SSH and SFTP, protect data, making it harder for attackers to intercept sensitive information. Overall, this hands-on experience gave me a much clearer understanding of how different network protocols work and how to spot important details in large amounts of traffic data.

Lab 4: Applying User Authentication and Access Controls

Lab 4 focused on setting up user authentication and access controls to manage security within a network. The main challenge was making sure that the correct permissions were assigned to users based on their roles without accidentally allowing unauthorized access. I also had to troubleshoot some issues when testing user access. First, I created user accounts and security groups in Active Directory and made sure each user had the right permissions. Then, I set up security settings for folders so that only the right groups could access certain files, such as HR, Manager (MGR), and Developer (DEV) files. After that, I

tested different user accounts to make sure the restrictions were working correctly. I also worked with TrueNAS to configure shared folder access and make sure users could only see what they were allowed to access. This lab showed me how important it is to follow the principle of least privilege, which means giving users access only to what they need. Testing different accounts helped confirm that the security settings were working correctly and preventing unauthorized access. Learning how to set up and check authentication settings gave me useful experience in managing security for an organization and reinforced the importance of properly controlling user access.

Lab 5: Using Encryption to Enhance Confidentiality and Integrity

Challenges: Understanding how public and private keys worked together to encrypt and decrypt files took some time. At first, it was confusing to keep track of which key belonged to which user. I had to be really careful with how I shared and stored keys. Verifying digital signatures also came with a learning curve. It was a hands-on way to understand how encryption is used in real scenarios. Process & Tools: I used GPG and Kleopatra to create and exchange keys. I encrypted and decrypted files using both symmetric and asymmetric encryption and signed a document to test digital signatures. I practiced verifying those signatures to confirm the sender and integrity. I also used hybrid encryption to combine the strengths of both methods. Each step helped reinforce how encryption works in the real world. Takeaways: This lab helped me understand how encryption protects information from unauthorized access. I learned how public key cryptography makes secure communication possible. I also saw how digital signatures confirm both identity and integrity. It made me appreciate the complexity and power of encryption in cybersecurity.

Lab 6: Assessing Common Attack Vectors

Challenges: This lab involved a lot of different types of attacks, and it wasn't always clear if they were working right away. Sometimes I had to reset the setup or rerun scripts to get proper results. It was a challenge to remember the correct steps for each attack type. I also had to make sure I didn't crash the system while simulating a DoS attack. It really pushed me to work carefully and think ahead. Process & Tools: I performed SQL injections, created malware payloads with msfvenom, launched DoS attacks, and built phishing emails. I also tested some of the attacks against live systems in a lab setting. After each attack, I analyzed how it worked and what could have prevented it. Then I looked into defensive tools like input validation, antivirus software, and employee training. I made notes on how each attack type could be detected or stopped. Takeaways: I saw how easy it can be for attackers to exploit weak systems and how important it is to train employees and use strong defenses. This lab helped me understand how common attacks work from the

inside. I learned how attackers think and how to defend against that mindset. It also made me more aware of how one vulnerability can lead to major issues.

Lab 7: Implementing an IT Security Policy

Challenges: Applying security policies across multiple systems was more complex than I expected. Some settings didn't stick right away and needed extra troubleshooting. I had to make sure everything from password rules to antivirus and mobile device security was correctly enforced. Getting policies to apply to mobile devices also required looking at settings I wasn't as familiar with. It took patience to double-check that every rule was active and doing what it should. **Process & Tools:** I configured password and antivirus policies using Active Directory, then used Microsoft's baseline settings to tighten security. I also applied settings on a mobile device, enabling encryption, screen locks, and app scanning. I tested features like Find My Device and reviewed policy enforcement logs. I researched acceptable use and privacy policies from real organizations to better understand how rules are created and applied. This gave me a good balance of technical skills and policy knowledge. **Takeaways:** This lab taught me that policies are just as important as tools when it comes to securing systems. I learned how to create rules that protect both devices and data. It also helped me understand that policies must be clear, enforceable, and tested regularly. Seeing the connection between mobile and desktop security made me appreciate the importance of consistency.

Lab 8: Implementing Security Monitoring and Logging

Challenges: Figuring out how to track down specific security events took some effort, especially on the Linux side. At first, I wasn't sure how to filter the logs or where to find failed login attempts. It also took time to configure Snort properly and make sure it detected traffic as expected. Understanding the different alerts and what they meant required going through documentation. The lab pushed me to improve my attention to detail and patience. **Process & Tools:** I used Event Viewer to look for failed Windows logins and rsyslog for Linux logs. I configured Snort to monitor traffic and set it up in both IDS and IPS modes. I used Tripwire to monitor file integrity and alert me when something changed. I analyzed the output from each tool and reviewed logs to track events. I also worked with ICMP traffic to generate alerts and understand normal vs. abnormal behavior. **Takeaways:** This lab showed me how important logging and monitoring are in detecting threats early. I learned how different tools work together to provide a complete view of system health and security. Being able to read logs and interpret alerts is a valuable skill in cybersecurity. I feel more confident now in my ability to detect potential issues before they become full-blown incidents.

Lab 9: Configuring Backup and Recovery Functions

Challenges: Setting up backup schedules and testing failover functionality was a little overwhelming at first. I wasn't sure how to simulate a failure or what to look for in the results. Getting both servers to load balance correctly took a few tries. I had to learn how to monitor traffic shifts and make sure everything was running smoothly. It was a good challenge that showed me how redundancy works in practice. **Process & Tools:** I used Windows Server Backup to create system state backups and schedule them at regular intervals. Then I configured HAProxy to distribute traffic between two servers and tested failover by taking one server offline. I verified the behavior by checking session logs and DNS responses. I also created NFS shares and confirmed data could still be accessed during a failure. The tools gave me a hands-on view of high availability setups. **Takeaways:** This lab helped me understand how backups and redundancy protect against data loss and downtime. It showed me that high availability isn't just about having two servers, it's about making sure they work together correctly. I gained practical experience in testing failovers and reading performance metrics. It made me feel more confident in my ability to plan for system recovery.

Lab 10: Performing Incident Response and Forensic Analysis

Challenges: There was a lot of information to go through when analyzing PCAP files and disk images. Trying to piece together what happened and when felt like solving a puzzle. Writing the incident report was also challenging because I needed to include all relevant details while staying clear and organized. It was tough balancing technical evidence with easy to understand explanations. I had to be extra careful not to miss anything important. **Process & Tools:** I used Wireshark to review PCAP files and searched disk images for keylogger evidence. I tracked timestamps and email messages that tied into the attack timeline. I documented everything in a structured incident response report, including who was involved, how the attack happened, and what systems were affected. I also looked for signs of spyware and documented red flags that pointed to phishing. Each step built up the full picture of the incident. **Takeaways:** This lab showed me what it takes to conduct a real forensic investigation. I learned how to trace events through log files, analyze evidence, and build a timeline. I also saw how important good documentation is when reporting an incident. Overall, it gave me a deeper appreciation for how security teams respond to and learn from breaches.

Lab 11: Ethical Hacking Part 1

Challenges: This lab was mostly self-guided, which made it feel more like a real-world challenge. I didn't have step-by-step instructions, so I had to rely on my knowledge to

explore the files and find weaknesses. It was difficult figuring out where to start and what tools would help. I had to be careful not to overlook small details that could lead to the flag. It was one of the most open ended labs we've done. Process & Tools: I analyzed files and system output that simulated a vulnerable environment. I looked for patterns, clues, and hidden content that would reveal the next steps. I applied different ethical hacking techniques to uncover weaknesses and tested commands cautiously. I kept notes on what worked and what didn't and documented my findings as if reporting them responsibly. This helped me build a hacker's mindset. Takeaways: This lab helped me understand how ethical hackers think and work through problems. It taught me to be persistent, creative, and cautious. I realized that not all vulnerabilities are obvious and that sometimes you have to dig deep. It was a good test of everything I've learned so far and showed me how to approach real challenges ethically and methodically